



HIGH COMPUTING POWER AND NATIVE WIFI/BLE COMMUNICATION



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Practical Thinking: Side Blinker Equipped with Diotec TVS Diode 5.0SMCJ48CA-AQ

Signal lights have an important safety function. They demonstrate deceleration and a change in direction of the vehicle and make the vehicle more visible to other road users when it is dark or during poor visibility. Any flashing light can be called a blinker but it usually refers to the turn signal on a car. Before you suddenly turn left, be sure to put your blinker on.

This is particularly important in automotive environments, where electrical fluctuations caused by alternators, switching loads or inductive elements can damage



electronic parts. By placing the TVS diode across the power lines or near the protected components the circuit remains resilient against transient events, ensuring the reliable operation of the side blinker system. Proper installation including correct polarity and voltage compatibility is crucial to maximizing the diode effectiveness in preventing electrical damage.

The side blinker or turn signal is an essential component in automotive lighting, typically operating within a 12V or 24V system. To protect the circuit from voltage spikes and transients, the Diotec TVS diode 5.0SMCJ48CA-AQ is integrated into the design. This bidirectional transient voltage suppression diode is specifically designed to handle overvoltage conditions, with a stand-off voltage of 48V and a breakdown voltage range of 53.3V to 58.9V. Given its clamping voltage of approximately 77.4V and peak pulse power of 1500W, the diode effectively absorbs sudden voltage surges, safeguarding sensitive components such as LEDs, relays or control circuits.



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